



NILUS LAB

Reversing Cardiac Ageing Without Oncogene Risk

Generative cellular reprogramming therapeutics targeting cardiomyopathy.
Powered by the Zenith v28.0 GOLD single-cell foundation model.

DEEP BIO ACCELERATOR - LAUNCH TRACK INTERVIEW PRESENTATION

CLINICAL RISK & LIMITS

Cardiac Senescence & Reprogramming Limits

- **64M+ Heart Failure Patients**
Epigenetic decay and loss of contractility drive global mortality.
- **Oncogenic Risk of OSKM Factors**
Classic Yamanaka factors trigger uncontrolled dedifferentiation.
- **Delivery & Specificity Barriers**
Current therapies lack safe, cardiac-specific lineage targets.

ZENITH PLATFORM SOLUTION

Safe, Target-Specific Rejuvenation

- **In-Silico Safety Gating**
Dual-threshold gates eliminate oncogenes and preserve lineage.
- **Trained on 500,000 Single Cells**
Maps cardiac manifolds via scVI variational autoencoders.
- **-11.9y Epigenetic Age Reversal**
Trained on real donor ages with a cross-validated MAE of 6.0y.

1 scVI Generative Core

- In-Silico Modeling & Simulation
- Trained on 500,000 human cells
- 4,908 variable gene dimensions
- 20-dimensional latent manifold
- HCA + PERIHEART datasets

2 Safety Auditing

- Oncogene Blacklist Check
- Screens out MYC / OCT4 factors
- Dedifferentiation limits gated
- Ensures cell identity stability
- Monitors TNNT2 / ACTN2 levels

3 AF3 & ElasticNet Clock

- AlphaFold 3 Valency predictions
- Models 3D TF-DNA docking
- ElasticNet age regression clock
- MAE = 6.0y on real HCA donors
- Transient expression windows

MARKET POSITIONING & COMPETITION

Target Market, Competition & Differentiation

- **CVD Longevity Market** - \$50B addressable cardiomyopathy indication target
- **Retro Bio & Altos Labs** - Broad systemic vectors. Nilus focuses on cardiac spe
- **Turn Biotechnologies** - Dermatology focus. Nilus is optimized for cardiac line
- **Wetlab Commercialization** - Bringing science out of the lab and into the market.

WETLAB VALIDATION FUNDING ASK

Raising \$1.5M Seed round specifically to fund cocktail synthesis, calcium imaging, ATAC-seq chromatin profiling, and rodent safety assays.

\$1.5M Seed

WETLAB FUNDING ASK

-11.9 Years

EPIGENETIC AGE RESET (MAE=6.0y)

THE TEAM

Team Chemistry & Dynamics

- **Computational VAE Core:**
Expertise in deep latent manifolds.
- **Molecular Biology Lab:**
Specialists in cell transfections.
- **Long-Term Synergy:**
Collaborated on single-cell research.
- **BIO 2026 Summit Target:**
Preparing to showcase wetlab data.

DEVELOPMENT ROADMAP

Key Execution Milestones

**Phase 1: Cocktail Synthesis & Wetlab Setup**

Synthesize target cocktails and setup primary cardiomyocyte models.

**Phase 2: Preclinical In-Vitro Assaying**

Calcium-transient imaging & chromatin accessibility (ATAC-seq) profiles.

**Phase 3: In-Vivo Rodent Assays & Seed Close**

Conclude rodent cardiomyopathy safety safety trials and close seed funding.